COMPLETION OF AN ETALE MORPHISM IS RIG-ETALE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-etale-gives-rig-etale; the proof environment immediately following it is its own proof; the context span delivers the section situation that 'in the situation above' refers to, and the template argument with its lemma substitution is stated explicitly in the proof

Editorial note. The label lemma-etale-gives-rig-etale names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the /T completion functor and its cartesian square (section-completion-functor setup, delivered as context)
- the template argument of lemma-smooth-gives-rig-smooth (same section), with the smoothness input lemmas lemma-rig-smooth and lemma-rig-smooth-morphisms replaced by their rig-etale counterparts lemma-rig-etale and lemma-rig-etale-morphisms
- etale morphisms are locally of finite type, and the hypotheses that X is locally Noetherian and f locally of finite type

Source reference. Original lemma 0APV: sources/stacks-chow/tags.txt, source lines 13488–13488. Prerequisite pointer invoked at 6244; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0ALQ: sources/stacks-restricted/pinned-original-source.tex, source lines 1981–1981. Prerequisite pointer invoked at 6197; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GCZ: sources/stacks-restricted/pinned-original-source.tex, source lines 5047–5047. Prerequisite pointer invoked at 6197; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAK: sources/stacks-restricted/pinned-original-source.tex, source lines 943–943. Prerequisite pointer invoked at 6195; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 0GCQ: sources/stacks-restricted/pinned-original-source.tex, source lines 4642–4642. Prerequisite pointer invoked at 6195; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GDH: sources/stacks-restricted/pinned-original-source.tex, source lines 6123–6123. Prerequisite pointer invoked at 6194; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

1. The completion functor

In this section we consider the following situation. First we fix a base scheme S. All rings, topological rings, schemes, algebraic spaces, and formal algebraic spaces and morphisms between these will be over S. Next, we fix an algebraic space X and a closed subset $T \subset |X|$. We denote $U \subset X$ be the open subspace with $|U| = |X| \setminus T$. Picture

$$U \to X \quad |X| = |U| \amalg T$$

In this situation, given an algebraic space X' over X, i.e., an algebraic space X' endowed with a morphism $f : X' \to X$, then we denote $T' \subset |X'|$ the inverse image of T and we let $U' \subset X'$ be the open subspace with $|U'| = |X'| \setminus T'$. Picture

$$U' = f^{-1}U \qquad \begin{array}{ccc} U' & \longrightarrow & X' \\ \downarrow & & \downarrow{\scriptstyle f} \\ U & \longrightarrow & X \end{array} \qquad \begin{array}{ccccc} |U'| & \longrightarrow & |X'| & \longleftarrow & T' \\ \downarrow & & \downarrow{\scriptstyle |f|} & & \downarrow \\ |U| & \longrightarrow & |X| & \longleftarrow & T \end{array} \qquad T' = |f|^{-1}T$$

We will relate properties of f to properties of the induced morphism

$$f_{/T} : X'_{/T'} \longrightarrow X_{/T}$$

of formal completions. As indicated in the displayed formula, we will denote this morphism $f_{/T}$. We have already seen that $f_{/T}$ is representable by algebraic spaces in Formal Spaces, Lemma 0APV. In fact, as the proof of that lemma shows, the diagram

$$\begin{array}{ccc} X'_{/T'} & \longrightarrow & X' \\ {\scriptstyle f_{/T}}\downarrow & & \downarrow{\scriptstyle f} \\ X_{/T} & \longrightarrow & X \end{array}$$

is cartesian. Please keep this fact in mind whilst reading the lemmas stated and proved below.

Lemma 1.1. *In the situation above. If X is locally Noetherian, f is locally of finite type, and $U' \to U$ is étale, then $f_{/T}$ is rig-étale.*

Proof. The proof is exactly the same as the proof of Lemma 0GDH except with Lemmas 0GAK and 0GCQ replaced by Lemmas 0ALQ and 0GCZ □